

Gopher Industries – Food Remedy API

SIT374 – Capstone Team Project (A)

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Ticket – RE006

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1. Introduction

Understanding how food energy is measured is fundamental to our work. Whether we are creating tracking functionalities, producing educational materials, or developing databases, a collective, fundamental understanding of what a “calorie” genuinely signifies guarantees precision and uniformity throughout our projects. This document acts as a foundational guide for the team, explaining the science of measurement and its practical use on food labels.

This ticket document emphasizes two main aspects: the scientific definition of calories as a unit of energy, and the standardized way this information is presented to customers. By establishing this common understanding, I guarantee that my features and communication are both scientifically valid and easy for users to understand.

2. What is a calorie?

A calorie is merely a measure of energy. In relation to food, it quantifies the energy delivered by a specific food or beverage when your body ingests and metabolizes it. This energy is crucial for all activities, from breathing and thinking to running and digestion.

The scientific definition:

- Technically, one calorie is the amount of energy needed to raise the temperature of 1 gram of water by 1 degree Celsius.
- Nonetheless, when discussing food, we are referring to kilocalories (kcal). One kilocalorie is equivalent to 1,000 calories. This is the reason you frequently see “calorie” with a capital “C” on food labels: it’s shorthand for kilocalories.

Think of your body like a car and food like fuel. Just as a car needs gasoline to run, your body needs energy from calories to function. Different foods provide different amounts of this fuel.

3. How calories appear on food labels

In many parts of the world food labeling regulations require that the energy content of a product is clearly displayed while the core information is consistent the way it is presented can vary slightly by region here is a comparison of these standards in UK and Australia.

- Kcal (kilocalories): the most common unit consumers are familiar with. When someone says, “I’m eating a 150-calories snack,” they mean 150 kcal.
- KJ (kilojoules): the metric unit of energy. In the UK, it is often displayed alongside kcal. In Australia, KJ is the primary unit of

measurement for energy on food labels. One kcal is equal to approximately 4.184 KJ.

Label feature	United Kingdoms (UK)	Australia
Primary energy unit	Kcal is the dominant unit shown to consumers, with KJ often displayed alongside.	KJ is the legislated primary unit. Kcal is sometime shown in brackets or may be absent.
Per 100g/100ml	Mandatory. Allows for easy comparison between different products.	Mandatory. This is the standard reference for comparing products.
Per serving/ portion	Mandatory. Shows the consumer how many calories are in a suggested serving size.	Mandatory. The declaration must include the energy per serving, which is defined by the manufacturer.
Additional context	May include % reference intake (RI), showing the contribution of a serving to an average adult's daily intake.	May include % daily intake (DI) based on an average adult diet. This serves the same purpose as the UK's %RI.

4. Daily recommended intake

Understanding daily calorie intake is important for interpreting food labels and making healthy dietary decisions. The required calorie intake varies depending on age, gender, and activity level.

1. Adults (Male vs Female)

- Adult male:

An average adult male requires approximately 2,500 kcal (10,500 KJ) per day to maintain a healthy weight.

- **Adult female:**

An average adult female requires approximately 2,000 kcal (8,400 KJ) per day.

These values are general guidelines and may vary depending on physical activity levels and individual metabolism.

2. Children

Children require fewer calories than adults because of their small body size and different energy needs.

- Children (4-8 years): 1,200 – 1,400 kcal/day
- Children (14-18 years): 1,800 – 2,400 kcal/day

The exact intake depends on growth, activity level, and gender.

5. Key takeaways & visual aid

Nutrition Information		
Servings per package – 16 Serving size – 30g (2/3 cup)		
	Per serve	Per 100g
Energy	432kJ	1441kJ
Protein	2.8g	9.3g
Fat		
Total	0.4g	1.2g
Saturated	0.1g	0.3g
Carbohydrate		
Total	18.9g	62.9g
Sugars	3.5g	11.8g
Fibre	6.4g	21.2g
Sodium	65mg	215mg
Ingredients: Cereals (76%) (wheat, oatbran, barley), psyllium husk (11%, sugar, rice, malt extract, honey, salt, vitamins.		
Ingredients ▲ Listed from greatest to smallest by weight. Use this to check the first three ingredients for items high in saturated fat, sodium (salt) or added sugar.		

Total Fat ▶
Generally choose foods with less than 10g per 100g.
For milk, yogurt and icecream, choose less than 2g per 100g.
For cheese, choose less than 15g per 100g.

Saturated Fat ▶
Aim for the lowest, per 100g.
Less than 3g per 100g is best.

Other names for ingredients high in saturated fat: Animal tallow, beef fat, butter, chocolate, milk solids, coconut, coconut oil/milk/cream, copha, cream, ghee, dripping, lard, suet, palm oil, sour cream, vegetable shortening.

Fibre ▶
Not all labels include fibre.
Choose breads and cereals with 3g or more per serve

◀ 100g Column and Serving Size
If comparing nutrients in similar food products use the per 100g column. If calculating how much of a nutrient, or how many kilojoules you will actually eat, use the per serve column. But check whether your portion size is the same as the serve size.

Energy
Check how many kJ per serve to decide how much is a serve of a 'discretionary' food, which has 600kJ per serve.

Sugars
Avoiding sugar completely is not necessary, but try to avoid larger amounts of added sugars. If sugar content per 100g is more than 15g, check that sugar (or alternative names for added sugar) is not listed high on the ingredient list.

Other names for added sugar: Dextrose, fructose, glucose, golden syrup, honey, maple syrup, sucrose, malt, maltose, lactose, brown sugar, caster sugar, maple syrup, raw sugar, sucrose.

◀ Sodium (Salt)
Choose lower sodium options among similar foods. Food with less than 400mg per 100g are good, and less than 120mg per 100g is best.

Other names for high salt ingredients: Baking powder, celery salt, garlic salt, meat/yeast extract, monosodium glutamate, (MSG), onion salt, rock salt, sea salt, sodium, sodium ascorbate, sodium bicarbonate, sodium nitrate/nitrite, stock cubes, vegetable salt.

Nutrition Facts	
4 servings per container	
Serving size 1 1/2 cup (208g)	
Amount per serving	
Calories	240
% Daily Value*	
Total Fat 4g	5%
Saturated Fat 1.5g	8%
Trans Fat 0g	
Cholesterol 5mg	2%
Sodium 430mg	19%
Total Carbohydrate 46g	17%
Dietary Fiber 7g	25%
Total Sugars 4g	
Includes 2g Added Sugars	4%
Protein 11g	
Vitamin D 2mcg	10%
Calcium 260mg	20%
Iron 6mg	35%
Potassium 240mg	6%
* The % Daily Value (DV) tells you how much a nutrient in a serving of food contributes to a daily diet. 2,000 calories a day is used for general nutrition advice.	

These label pictures show the proper information About the food It shows the regional difference and what it means for the user.

- KJ is a scientific metric measurement (1 kcal = 4.184 KJ). This is the primary number in Australia and in UK, “kcal” is the primary number user’s track
- Per 100g is the best way to compare products both in Australia and UK.
- Per serving is the best way to log what you eat, it shows the actual energy in one portion.

6. Conclusion & next steps

This overview establishes a clear, shared definition of a calories and explains its presentation on food labels. Understanding that a calorie is a unit of energy, that “Kcal” is the consumer-friendly term for kilocalories, and how to interpret the different parts of a label (per 100g vs. per serving) is the foundation for many of our product decisions.

Crucially, recognizing the regional differences such as Australia’s primary use of kilojoules (KJ) compared to the UK’s focus on kilocalories (kcal) ensures we can build a product that is accurate and intuitive for users in multiple markets. This shared knowledge helps us create features, write copies, and analyze competitors from a position of confidence and accuracy, building a product that is both engaging and trustworthy.

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